# Mid-Term Examination (Afternoon)

## Model Answers

(a) $x(t) = \cos(650\pi t) + 2\sin(720\pi t)$

$f_s = 500\text{Hz}$; $\frac{f_s}{2} = 250\text{ Hz}$

$f_1 = 325\text{ Hz}$

$f_1 > \frac{f_s}{2}$

This frequency is affected by aliasing

$f_2 = 360\text{ Hz}$

$f_2 > \frac{f_s}{2}$

affected by aliasing

[2]

∴ aliased frequency

$f_1' = f_1 - k f_s$

$= 325 - 1 \cdot 500 = -175\text{ Hz}$

∴ aliased frequency

$f_2' = f_2 - 1 \cdot f_s$

$= 360 - 500 = -140\text{ Hz}$

The resulting frequencies in the discrete-time signal are:

$f_1' = -175\text{ Hz}$ & $f_2' = 140\text{ Hz}$

The discrete time signal is;

$$x[n] = \cos\left(2\pi\left(\frac{-175}{500}\right)n\right) + 2\sin\left(2\pi\left(\frac{-140}{500}\right)n\right)$$

[2]

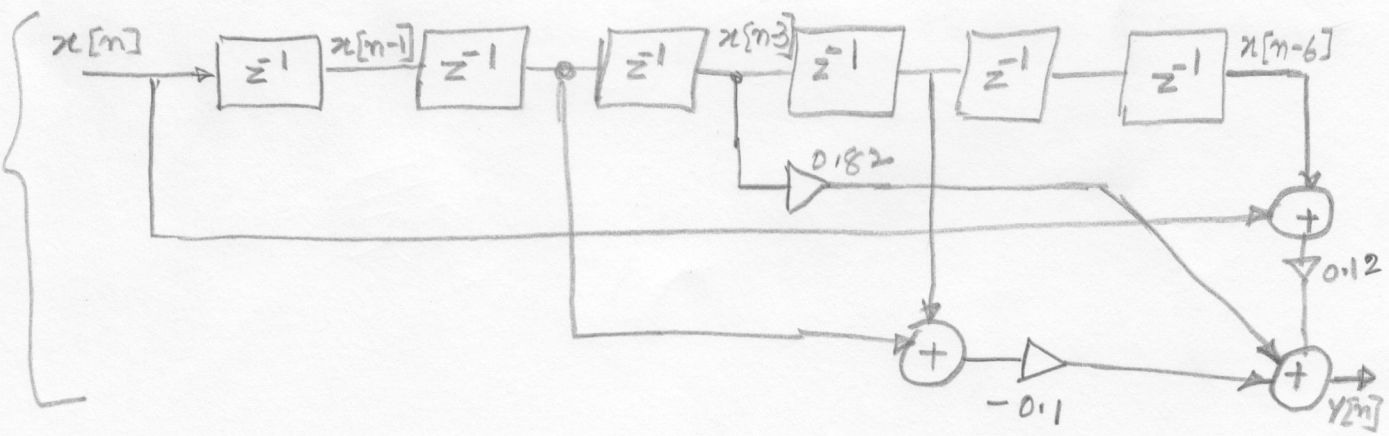
$$\boxed{x[n] = \cos\left[\frac{7\pi}{10}n\right] - 2\sin\left[\frac{14\pi}{25}n\right]}$$

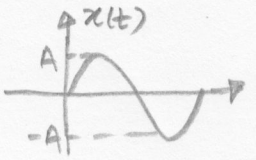
(b) (i) $h[n] = \{0.12, 0, -0.1, 0.82, -0.1, 0, 0.12\}$ $h[n] = 0$ $n \geq 7$

(ii) $y[n] = 0.12\{x[n] + x[n-6]\} - 0.1\{x[n-2] + x[n-4]\} + 0.82\,x[n-3]$

[3]

[3]



(c) 


step size $\Delta V = \frac{2A}{N-1}$

$$SQNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{\left(\frac{A}{\sqrt{2}}\right)^2}{\frac{\Delta V^2}{12}}$$

$$= \frac{6A^2}{\Delta V^2} = \frac{6A^2}{\frac{4A^2}{(N-1)^2}} = \frac{3}{2}(N-1)^2$$ [2]

$$SQNR(dB) = 10 \log_{10} \frac{3}{2}(N-1)^2$$

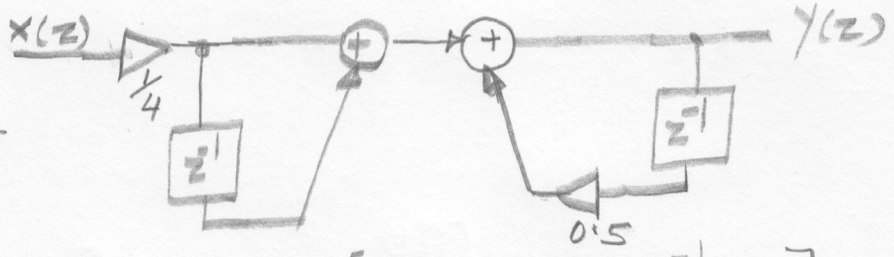
$$= 10 \log_{10} \frac{3}{2} + 20 \log_{10} (N-1)$$

$$\boxed{SQNR(dB) = 1.76 + 20 \log_{10} (N-1)}$$ [1]

SoLO - level 2:

$Q_2$ (a) (i) $H(z) = \frac{1}{4}(1+z^{-1}) \cdot \frac{1}{1-0.5z^{-1}}$

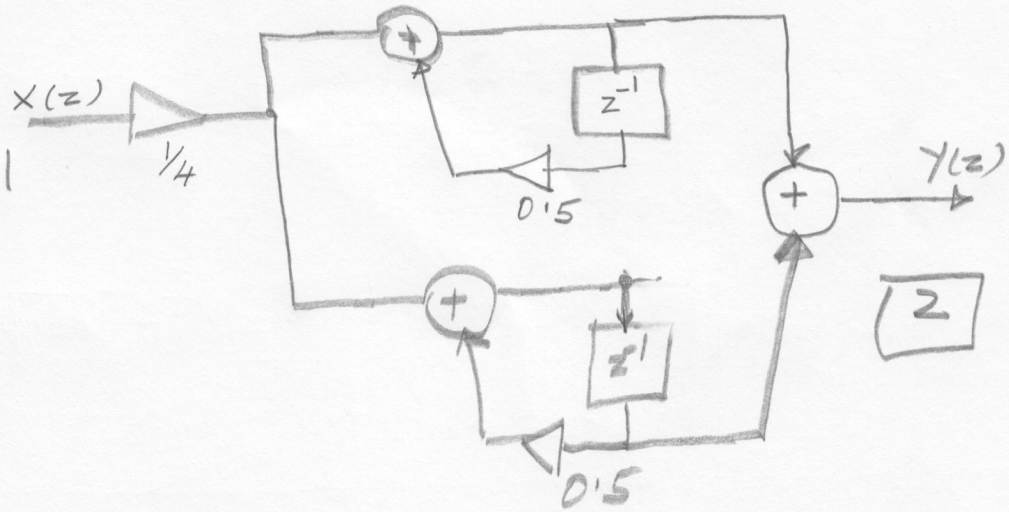
Cascade




[2]

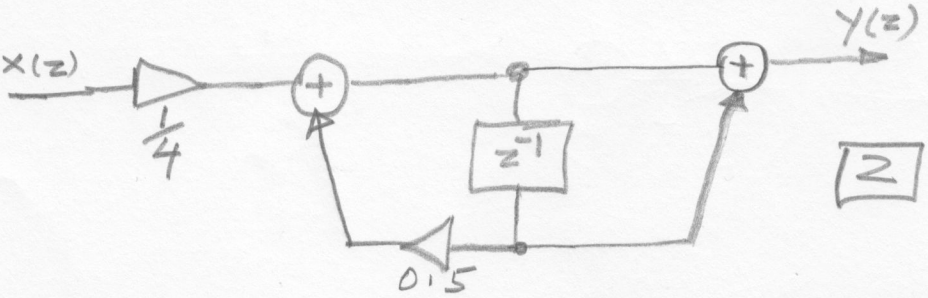
(ii) $H(z) = \frac{1}{4}\left[\frac{1}{1-0.5z^{-1}} + \frac{z^{-1}}{1-0.5z^{-1}}\right]$ [1]

Parallel



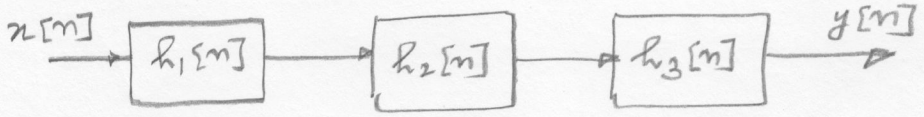

[2]

Canonic




[2]

(b)




1

$h_c[n] = h_1[n] * h_2[n] * h_3[n]$

$h_c[n] = \{\delta[n] - \delta[n-1]\} * \{\delta[n] - \delta[n-1]\} * u[n]$

$H_c(z) = (1 - z^{-1}) \cdot (1 - z^{-1}) \cdot \frac{1}{1 - z^{-1}}$ 2

$H_c(z) = 1 - z^{-1}$

$\boxed{h_c[n] = \delta[n] - \delta[n-1]}$

alternatively

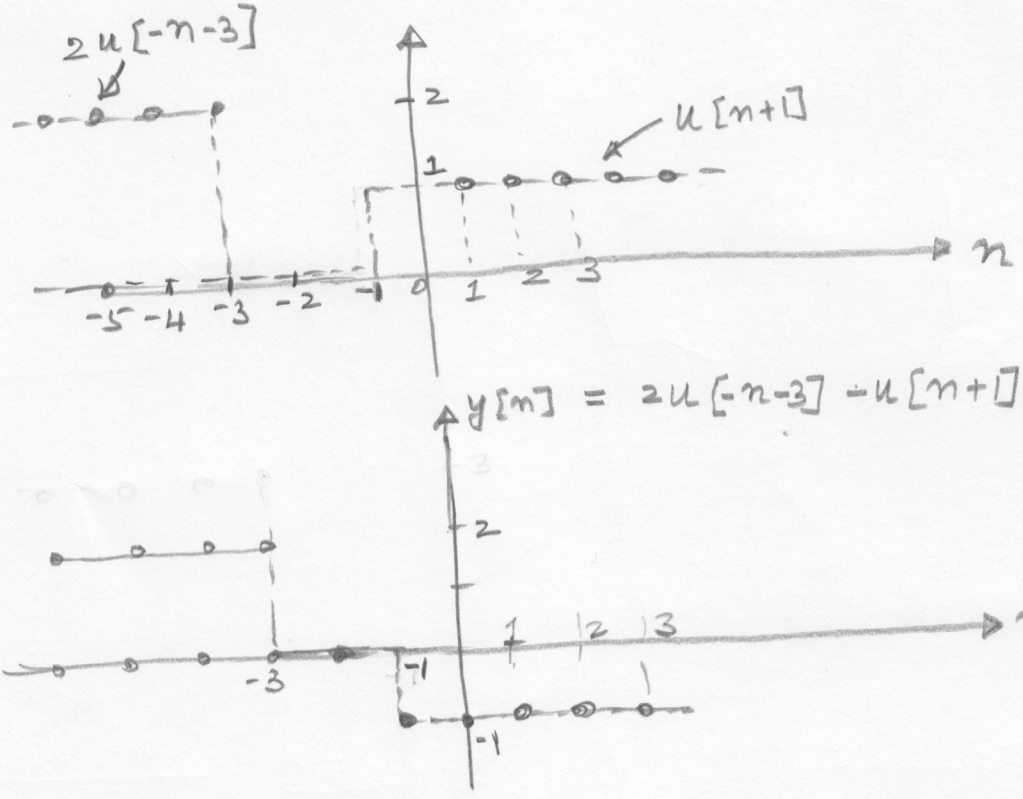
$h_c[n] = \{\delta[n] * \delta[n-1]\} * \{u[n] - u[n-1]\}$

$u[n] - u[n-1] = \delta[n]$

$= \{\delta[n] - \delta[n-1]\} * \delta[n]$

$\boxed{h_c[n] = \delta[n] - \delta[n-1]}$

(c)




3

SOLO - Level 2

Q3 (a) $H(z) = \frac{1}{3}\,\frac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$

$H(\theta) = \frac{1}{3}\,\frac{1+\frac{1}{2}e^{-j\theta}}{1-\frac{1}{2}e^{-j\theta}}$ [1]

$|H(\theta)|^2 = H(\theta)\cdot H^*(\theta) = \left(\frac{1}{3}\right)^2 \frac{1+\frac{1}{2}e^{-j\theta}}{1-\frac{1}{2}e^{-j\theta}} \cdot \frac{1+\frac{1}{2}e^{j\theta}}{1-\frac{1}{2}e^{j\theta}}$

$= \frac{1}{9}\,\frac{\frac{5}{4}+\cos\theta}{\frac{5}{4}-\cos\theta}$ [2]

[2]

3 dB cut-off occurs at $\theta = \theta_c$, $|H(\theta)|^2 = \frac{1}{2}$

$\frac{1}{2} = \frac{1}{9}\,\frac{\frac{5}{4}+\cos\theta_c}{\frac{5}{4}-\cos\theta_c}$

$\frac{45}{4} - 9\cos\theta_c = \frac{5}{2} + 2\cos\theta_c$

$11\cos\theta_c = \frac{35}{4} \Rightarrow \boxed{\theta_c = \cos^{-1}\left(\frac{35}{44}\right) = 0.651 \text{ rad}}$

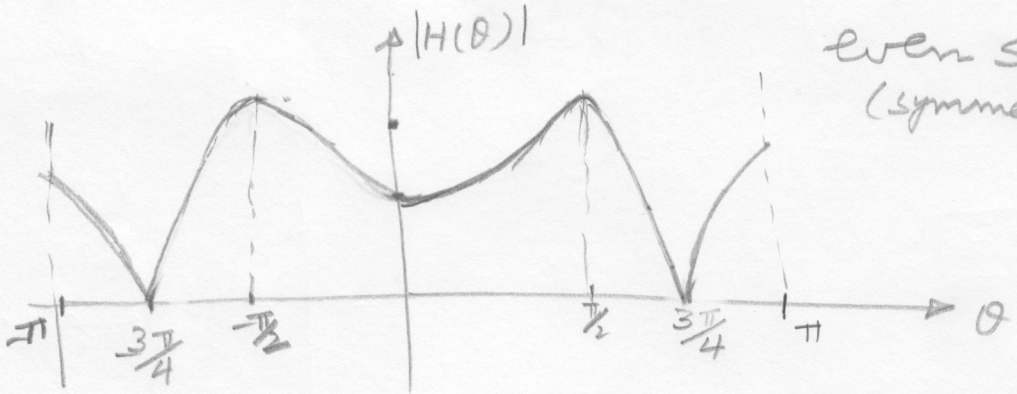
(b) $H(z) = \frac{(z-1\cdot e^{j\frac{3\pi}{4}})(z-1e^{-j\frac{3\pi}{4}})}{(z+\frac{1}{2}j)(z-\frac{1}{2}j)} = \frac{z^2-2\cos\frac{3\pi}{4}z+1}{z^2+\frac{1}{4}}$

[1] [1]

$= \frac{z^2-\sqrt{2}z+1}{z^2+\frac{1}{4}}$

[1] [2]

(c)




even spectrum (symmetrical)

[4]

SOLO - Level 2:

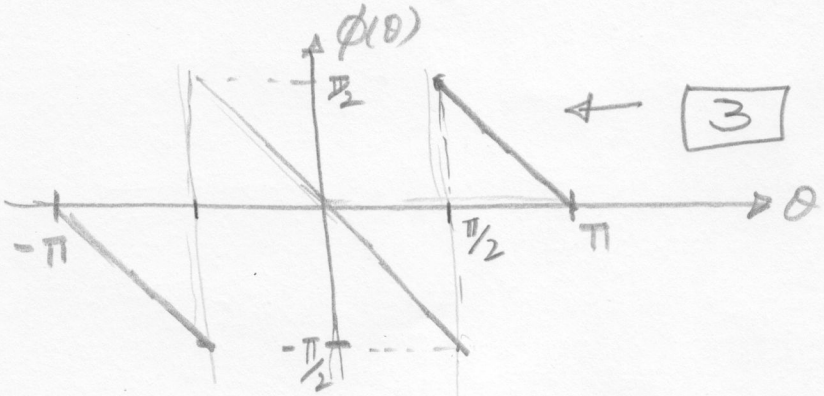
$Q_4$ (a) $\quad y(z) = x(z) + z^{-2}x(z)$

$H(z) = 1 + z^{-2} \quad \therefore H(\theta) = 1 + e^{-j2\theta} = e^{-j\theta}(e^{j\theta} + e^{-j\theta})$

$\therefore H(\theta) = e^{-j\theta} \underbrace{2\cos\theta}_{B(\theta)}$ [1]

$\phi(\theta) = -\theta \quad$ if $B(\theta) \geq 0$

$= -\theta \pm \pi \quad$ if $B(\theta) < 0$ [1]




[3]

(b) Frequency resolution $\Delta f = \frac{f_s}{N} = \frac{40000}{128} = 312.5\,Hz$

↑ Spacing between DFT values in Hz

Duration = no of samples × sampling Period

$= 128 \times \frac{1}{40\,kHz} = 3.2\,ms$ [2]

$f_0 = 0$; $f_1 = 312.5\,Hz$, $f_2 = 2 \times 312.5$; $f_3 = 3 \times 312.5$, $\cdots f_k = k \times 312.5$ [1]

$\therefore k = \frac{f_k}{312.5} = \frac{5 \times 10^3}{312.5} = 16$ [2]

At DFT indices of $k = 16$, a peak in the spectrum will appear.

Q.4 (c) $H(z) = \dfrac{1 - \frac{1}{a} z^{-1}}{1 - a z^{-1}} = \dfrac{1 - 2 z^{-1}}{1 - \frac{1}{2} z^{-1}}$ [1]

[2]

$$H(\theta) = \frac{1 - 2 e^{-j\theta}}{1 - \frac{1}{2} e^{-j\theta}} = \frac{(1 - 2\cos\theta) + j\, 2\sin\theta}{(1 - \frac{1}{2}\cos\theta) + j\, \frac{1}{2}\sin\theta}$$

$$|H(\theta)|^2 = \frac{(1 - 2\cos\theta)^2 + 4\sin^2\theta}{(1 - \frac{1}{2}\cos\theta)^2 + \frac{1}{4}\sin^2\theta} = \frac{5 - 4\cos\theta}{\frac{5}{4} - \cos\theta}$$

$= 4$ all-pass filter.

SOLO - Level 3

(a) (i) Note that the magnitude is constant for all $\theta$ and the phase is $-\pi/2$ for $0 \le \theta \le \pi$ and $\pi/2$ for $-\pi \le \theta < 0$

(2)
$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\theta) e^{jn\theta} d\theta$$

$$= \frac{1}{2\pi} \int_{-\pi}^{0} j e^{jn\theta} d\theta - \frac{1}{2\pi} \int_{0}^{\pi} j e^{jn\theta} d\theta$$

(4)
$$= \frac{1}{2\pi} \left[ \frac{j e^{jn\theta}}{jn} \right]_{-\pi}^{0} - \frac{1}{2\pi} \left[ \frac{j e^{jn\theta}}{jn} \right]_{0}^{\pi}$$

$$= \frac{1}{2\pi n} \left[ 1 - e^{-jn\pi} \right] - \frac{1}{2\pi n} \left[ e^{jn\pi} - 1 \right]$$

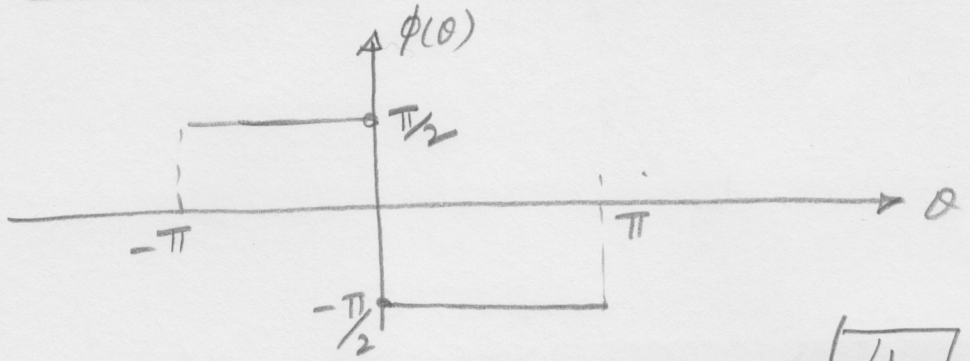
$$= \frac{1}{2\pi n} \left[ 2 - \left( e^{jn\pi} + e^{-jn\pi} \right) \right]$$

$$= \frac{1}{\pi n} \left[ 1 + \cos n\pi \right]$$

(2)
$$h[n] = \frac{1}{\pi n} \left[ 1 - (-1)^n \right]$$

(4)
$$\therefore h[n] = \begin{cases} \frac{2}{n\pi} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

(ii)




(4)

(b) (i) $h[n] = \delta[n] - \alpha\,\delta[n-1]$

$H(z) = 1 - \alpha z^{-1}$ ← [1]

$H(\theta) = 1 - \alpha e^{-j\theta} = (1-\alpha\cos\theta) + j\alpha\sin\theta$ ] [2]

[2] $\therefore |H(\theta)|^2 = (1-\alpha\cos\theta)^2 + (\alpha\sin\theta)^2 = 1 + \alpha^2 - 2\alpha\cos\theta$

↑ Magnitude squared

[2] The phase $\phi(\theta) = \tan^{-1}\dfrac{\alpha\sin\theta}{1-\alpha\cos\theta}$

(ii) [1] $\tau_g(\theta) = -\dfrac{d\phi(\theta)}{d\theta}$

↑ group delay

$= -\dfrac{d}{d\theta}\left(\tan^{-1}\dfrac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)$ ← $\left[\dfrac{d}{dx}\tan^{-1}u = \dfrac{1}{1+u^2}\dfrac{du}{dx}\right]$

[3]
$$= \frac{-1}{1+\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)^2} \cdot \frac{d}{d\theta}\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)$$

$$= \frac{-1}{1+\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)^2} \cdot \frac{(1-\alpha\cos\theta)\alpha\cos\theta - (\alpha\sin\theta)^2}{(1-\alpha\cos\theta)^2}$$

[1]
$$= -\frac{(1-\alpha\cos\theta)\alpha\cos\theta - (\alpha\sin\theta)^2}{(1-\alpha\cos\theta)^2 + (\alpha\sin\theta)^2}$$

$$= (-1)\frac{\alpha\cos\theta - \alpha^2\cos^2\theta - \alpha^2\sin^2\theta}{1+\alpha^2-2\alpha\cos\theta}$$

$$\tau_g = \frac{\alpha^2 - \alpha\cos\theta}{1+\alpha^2-2\alpha\cos\theta}$$

## Solo - Level H

(a) $H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2}$

$H(\theta) = b_0 + b_1 e^{-j\theta} + b_2 e^{-j2\theta}$

$H(\theta)\big|_{\theta=2\pi/3} = 0 = b_0 + b_1 e^{-j\frac{2\pi}{3}} + b_2 e^{-j2\times\frac{2\pi}{3}}$ — (1) [1]

↑ rejects the component at $\theta = 2\pi/3$

We have $|H(0)| = 1 \Rightarrow b_0 + b_1 + b_2 = 1$ — (2) [1]

For linear phase $b_0 = \pm b_2$ ; select $\boxed{b_0 = b_2}$ [2]

(otherwise $b_1 = 0$) $\therefore \boxed{2b_0 + b_1 = 1}$

From (1) $e^{-j\frac{2\pi}{3}}\left[b_0 e^{j\frac{2\pi}{3}} + b_0 e^{-j\frac{2\pi}{3}} + b_1\right] = 0$

$\therefore 2b_0 \underbrace{\cos\tfrac{2\pi}{3}}_{=-1/2} + b_1 = 0$

$\therefore \boxed{b_0 = b_1}$

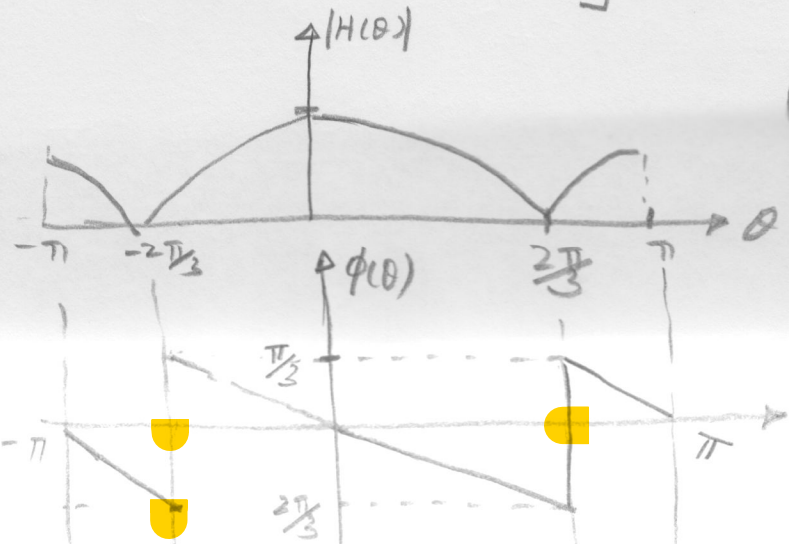
$\therefore b_0 = b_1 = b_2 = \frac{1}{3}$ (from 2) [1]

$\therefore H(\theta) = \frac{1}{3} e^{-j\theta}\underbrace{(1 + 2\cos\theta)}_{=B(\theta)}$ $\{b_0 = b_1 = b_2 = 1/3\}$ [1]

$|H(\theta)| = \frac{1}{3}(1 + 2\cos\theta)$

$\phi(\theta) = -\theta \quad B(\theta) \geq 0$

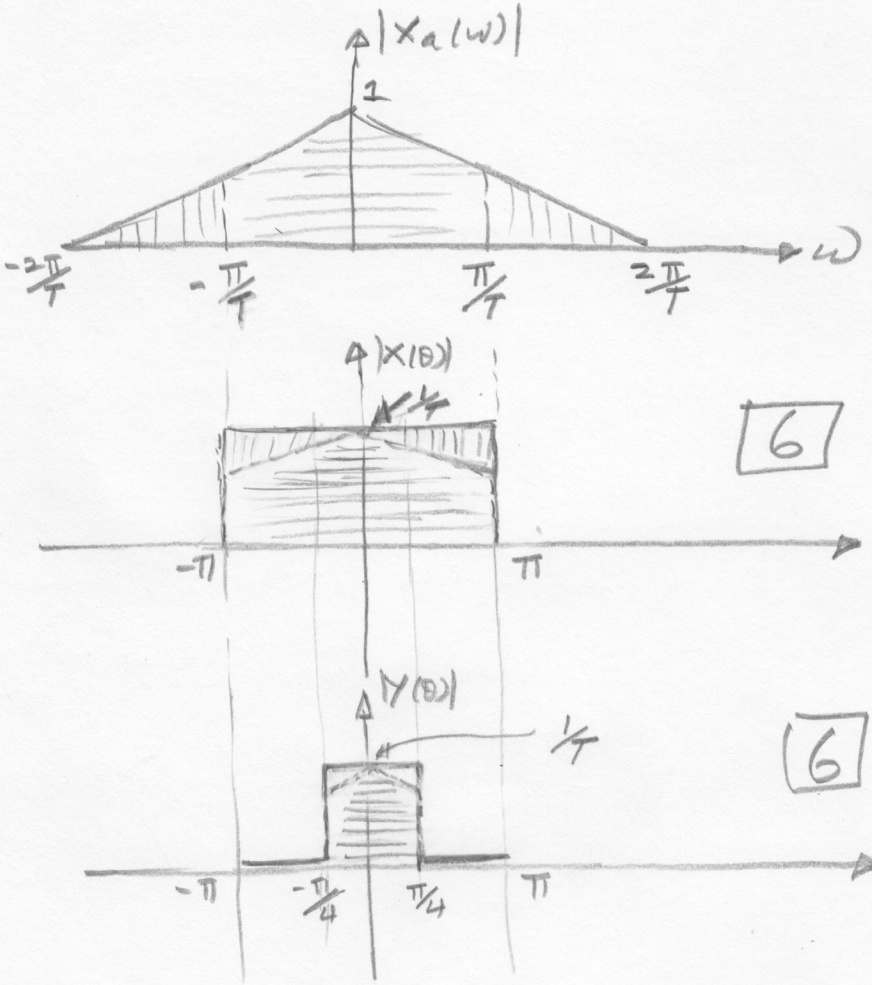
$= -\theta \pm \pi \quad B(\theta) < 0$ [1]




[2]

[2]

(c)

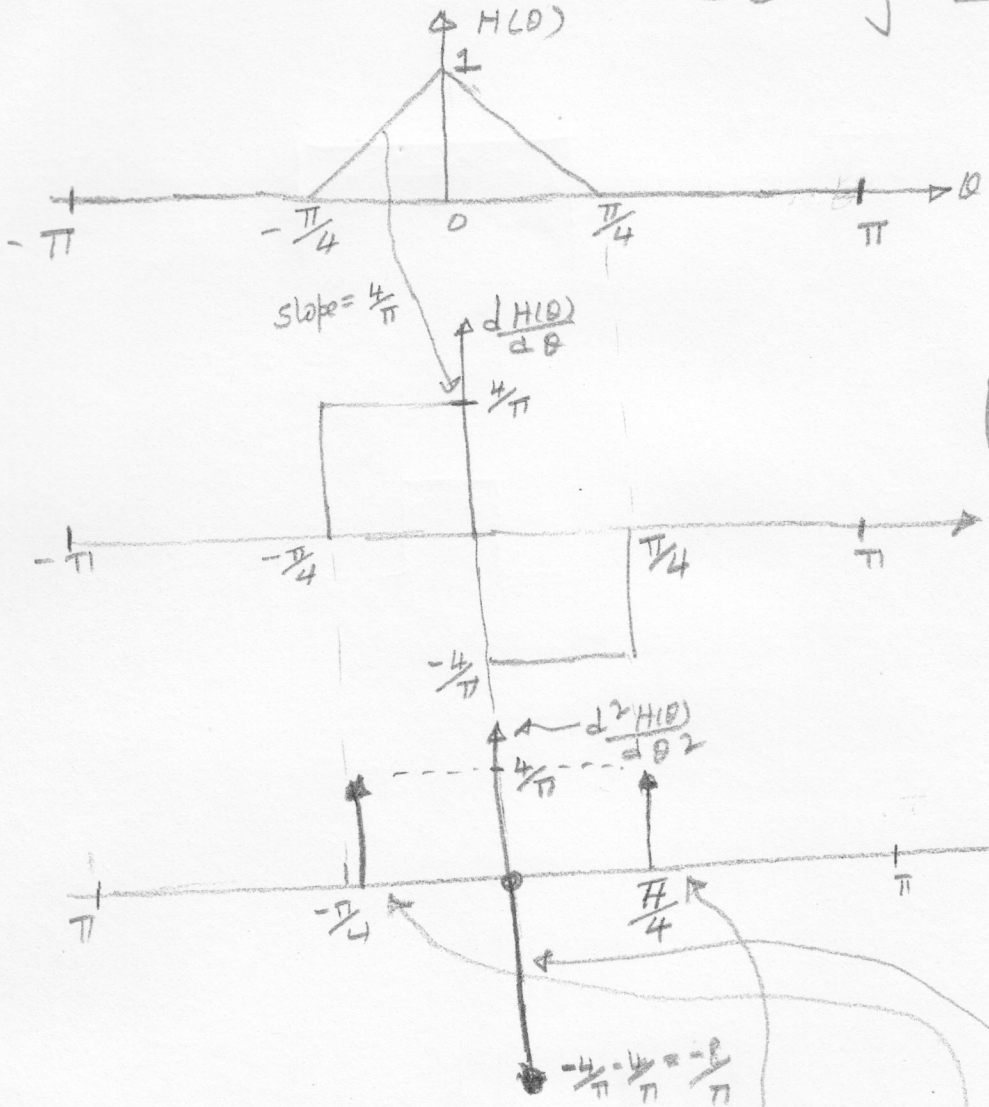



6

6

(a) To find the impulse response, we take IDTFT of $X(\theta)$. We can use differentiation property of DTFT

$$n\,h[n] \overset{DTFT}{\Longleftrightarrow} j\frac{dH(\theta)}{d\theta} \quad \} \boxed{1}$$




[2] { Taking IDTFT of $\frac{d^2H(\theta)}{d\theta^2} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{d^2H(\theta)}{d\theta^2} e^{jn\theta}\,d\theta$

$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left\{\frac{4}{\pi}\left[\delta(\theta-\frac{\pi}{4}) + \delta(\theta+\frac{\pi}{4})\right] - \frac{8}{\pi}\delta[\theta]\right\} e^{jn\theta}\,d\theta$

[1] { $= \frac{1}{2\pi}\left\{\frac{4}{\pi}\left(e^{jn\frac{\pi}{4}} + e^{-jn\frac{\pi}{4}}\right) - \frac{8}{\pi}\left(\frac{e^{jn\pi} - e^{-jn\pi}}{jn}\right)\right\} = \frac{4}{\pi^2}\cos\frac{n\pi}{4} - \frac{8}{\pi}\frac{\sin n\pi}{n\pi}$ ($\frac{\sin n\pi}{n\pi} = 1$)

$= \frac{4}{\pi^2}\cos\frac{n\pi}{4} - \frac{8}{\pi}$

[2] { To get $h[n]$ multiply by $\frac{1}{jn}\cdot\frac{1}{jn} = \frac{1}{n^2}$ $\left\{n\,h[n] \Longleftrightarrow j\frac{dH(\theta)}{d\theta}\right.$

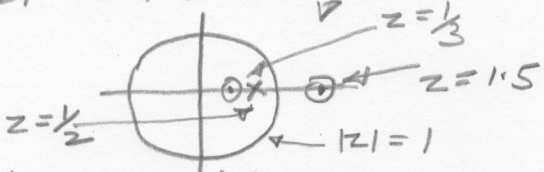
$$\boxed{h[n] = \frac{8}{\pi n^2} - \frac{4}{\pi^2 n^2}\cos\frac{n\pi}{4}}$$

(b) $H(z) = H_{min}(z) \cdot H_{LP}$ $\qquad H(z) = \dfrac{(3+z^{-1})(2-3z^{-1})}{1-\frac{1}{2}z^{-1}}$

This system is not minimum phase because the factor $(2-3z^{-1})$ corresponds to a zero outside the unit circle at $z = \frac{3}{2}$.

First, we reflect the zero about the unit circle and replace it with a pole:

[2] $\left\{ \dfrac{1}{2z^{-1}-3} \right.$



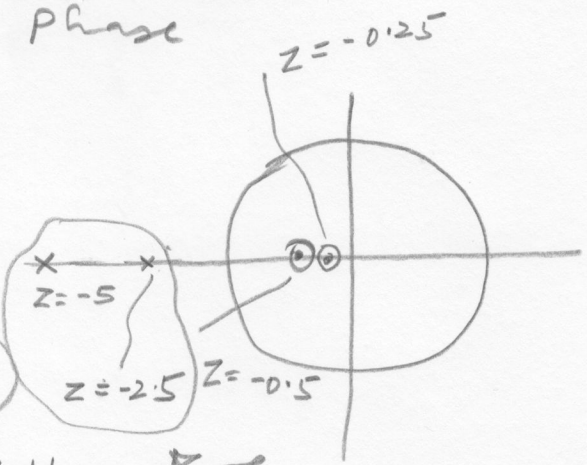

Then we multiply this term with the linear phase factor that has a zero at $z = \frac{2}{3}$ and a zero at $z = \frac{3}{2}$

[3] $\left\{ \dfrac{1}{2z^{-1}-3} \times (2z^{-1}-3)(2-3z^{-1}) \right.$

Thus the factorisation for $H(z)$ is

[3] $\left\{ H(z) = \dfrac{3+z^{-1}}{\underbrace{(1-\frac{1}{2}z^{-1})(2z^{-1}-3)}_{\text{minimum phase}}} \cdot \underbrace{(2z^{-1}-3)(2-3z^{-1})}_{\text{Linear Phase}} \right.$

(c) $H(z) = \dfrac{(z+0.5)(2z+0.5)}{(z+5)(2z+5)}$




(ii) $H_s(z) = H(z) \cdot A(z)$

[4] $\left\{ = \dfrac{(z+0.5)(2z+0.5)}{(z+5)(2z+5)} \cdot \underbrace{\dfrac{z+5}{1+5z}}_{\text{all-pass filters}} \cdot \underbrace{\dfrac{2z+5}{2+5z}}_{\text{all-pass filters}} \right.$ ← These two poles are outside the unit circle

[4] (i) $\therefore A(z) = \dfrac{z+5}{1+5z} \cdot \dfrac{2z+5}{2+5z}$

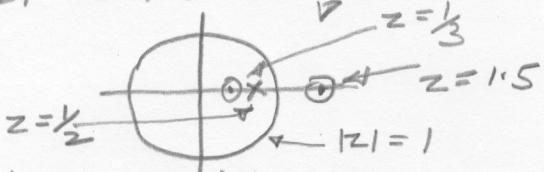
[3] (iii) $H_s(z)$ is causal & non-min phase

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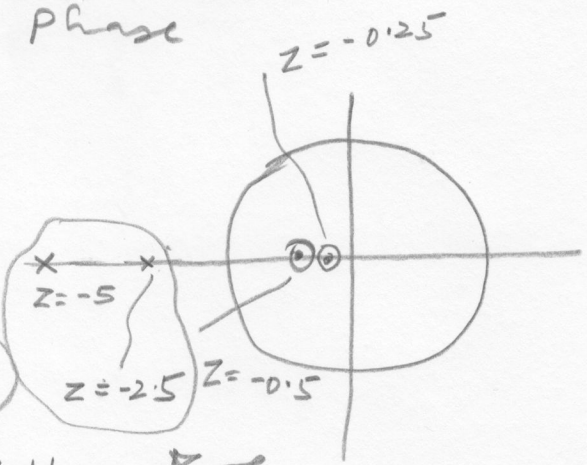

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Thus the factorisation for $H(z)$ is

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(c) $H(z) = \dfrac{(z+0.5)(2z+0.5)}{(z+5)(2z+5)}$




(ii) $H_s(z) = H(z) \cdot A(z)$

[4] $\left\{ = \dfrac{(z+0.5)(2z+0.5)}{(z+5)(2z+5)} \cdot \underbrace{\dfrac{z+5}{1+5z}}_{\text{all-pass filters}} \cdot \underbrace{\dfrac{2z+5}{2+5z}}_{\text{all-pass filters}} \right.$ ← These two poles are outside the unit circle

[4] (i) $\therefore A(z) = \dfrac{z+5}{1+5z} \cdot \dfrac{2z+5}{2+5z}$

[3] (iii) $H_s(z)$ is causal & non-min phase